

PEI-US 2016 Dataset — Detailed Knowledge Guide

Perceptions of Electoral Integrity: United States 2016

1. Purpose of This Document

This document is designed specifically as a knowledge source for the RAG component of an AI-powered analytics assistant. It provides background information about the PEI-US 2016 dataset, the meaning of electoral integrity, the survey methodology, the PEI Index, and important interpretation rules. It should be used for questions about definitions, concepts, methodology, and the meaning of the dataset. Numerical questions about the CSV should normally be handled by the Analytics Agent using Pandas rather than by guessing from this document.

2. Dataset Background

The dataset comes from the Electoral Integrity Project (EIP) study of the 2016 United States presidential election. The study is part of the broader Perceptions of Electoral Integrity (PEI) research program. The purpose of the project is to evaluate how experts perceive the quality and integrity of elections across different countries and subnational jurisdictions.

For the United States 2016 study, the analysis covered all 50 states and Washington, D.C. The survey collected 726 responses from election experts. The study is therefore an expert-based assessment of election quality rather than a database of individual voter behavior or a direct count of election violations.

3. What Does 'Electoral Integrity' Mean?

Electoral integrity refers broadly to whether an election is conducted according to accepted standards and procedures throughout the electoral cycle. The concept is broader than simply asking whether votes were counted correctly. It can include the legal framework, voter registration, candidate and party processes, campaigning, media, voting, counting, and the reporting of results.

The PEI framework therefore evaluates multiple parts of an election instead of treating electoral integrity as a single event occurring on election day.

4. Perceptions of Electoral Integrity (PEI) Index

The Perceptions of Electoral Integrity Index is a standardized measure created from expert survey responses. In the U.S. 2016 study, the core survey contains 49 items. These items are combined to produce an overall measure on a 0–100 scale. Higher scores generally indicate more positive expert perceptions of electoral integrity, while lower scores indicate less positive perceptions.

The PEI Index should be interpreted as a perception-based measure. A high or low PEI score does not by itself prove that an election was free from fraud or that fraud occurred. It represents the assessments of the experts who participated in the survey.

5. The 11 Stages of the Electoral Cycle

The PEI framework organizes electoral integrity into stages covering the wider electoral process. The stages are intended to help researchers identify which part of an election experts consider strong or weak. Depending on the

dataset/version, the corresponding variables may appear as stage-specific scores or individual survey indicators.

6. Core Survey Indicators

The 49 core items measure different aspects of electoral integrity. They are based on expert assessments rather than simple administrative counts. The questionnaire uses agreement/disagreement style response categories for the core items. These responses are processed into standardized indicators and aggregate indices.

Examples of concepts represented in the PEI framework include the quality of electoral laws, voter registration, electoral procedures, campaign-related conditions, media coverage, voting, counting, and the reporting of results. Individual columns in the supplied CSV should be matched to their exact definitions in the official codebook whenever possible.

7. What the Rows and Columns Represent

The exact structure of the CSV depends on the dataset release used in this project. In general, rows represent survey observations/respondents or aggregated observations, while columns represent variables describing respondent characteristics, electoral perceptions, or calculated measures.

Examples of variables encountered in this project include State, Age Range, Political Scale, National Electoral Integrity, and State Electoral Integrity. A column such as State can be used for grouping. A numerical integrity column can be used for averages, minimums, maximums, correlations, and other statistical analysis.

8. Difference Between Analytics and RAG

Analytics: Use the CSV and Pandas for questions requiring calculations. Examples include “What is the average National Electoral Integrity?”, “Which state has the highest average score?”, or “How many respondents are in each state?”

RAG: Use documentation for questions about meaning and methodology. Examples include “What is the PEI Index?”, “How was the survey conducted?”, “What does electoral integrity mean?”, and “What does a higher PEI score represent?”

Hybrid questions: Some questions require both. For example, “Which state has the highest electoral integrity and what does the score measure?” requires an analytical calculation followed by a definition from the RAG knowledge base.

9. Important Interpretation Rules

1. PEI is a perception measure. It should not be described as a direct measurement of fraud.
2. A higher score generally represents more positive expert assessments of electoral integrity.
3. Correlation between two variables does not prove that one caused the other.
4. Missing values should not automatically be converted to zero.
5. State comparisons should consider the number of observations available for each state.
6. The exact meaning of a variable should be taken from its official codebook when available.
7. Results should not be generalized beyond what the survey and dataset support.

10. Example Questions the RAG Agent Should Answer

Question	Expected knowledge
What is the PEI Index?	Definition of the Perceptions of Electoral Integrity Index.
What does a higher PEI score mean?	More positive expert perceptions of electoral integrity.
How many experts participated?	726 responses in the U.S. 2016 study.
What geographic areas were covered?	All 50 states and Washington, D.C.
What are the core indicators?	The study uses 49 core survey items.
What is electoral integrity?	Quality of the electoral process across the electoral cycle.
Is PEI a direct measure of fraud?	No. It is an expert-perception measure.

11. Example Analytical Questions

The Analytics Agent can answer questions such as:

- What is the average National Electoral Integrity?
- Which state has the highest average National Electoral Integrity?
- Which state has the lowest average score?
- How many respondents are there in each state?
- What is the median State Electoral Integrity?
- What is the correlation between two numerical integrity measures?
- How does the average score differ between groups?

12. Example Hybrid Questions

The final system should eventually support questions such as:

- “Which state has the highest National Electoral Integrity, and what does this measure mean?”
- “What is the average PEI score, and how should I interpret that score?”
- “Which electoral stage appears weakest in the data, and what does that stage represent?”

These questions require the agent to combine structured analysis with retrieved documentation.

13. How This Fits Our Project Architecture

Our planned system has two complementary paths. The Analytics path loads the CSV into Pandas and uses controlled tools for calculations. The RAG path loads documentation, splits it into chunks, creates embeddings, stores them in Chroma, and retrieves relevant passages. LangGraph will later route a user's question to the appropriate path and can combine both paths for hybrid questions.

Analytics path: User → LangGraph → Analytics Agent → Analytics Tool → Pandas → Result.

RAG path: User → LangGraph → Retriever → Relevant document chunks → Groq → Grounded answer.

Hybrid path: User → LangGraph → Analytics + Retriever → Groq → Combined answer.

14. RAG Metadata

When chunks are stored in Chroma, metadata such as the source filename and page number should be preserved. This allows the final application to identify where retrieved information came from and makes it easier to show sources in the final answer.

15. Limitations of This Guide

This document is a project-oriented summary and is not a replacement for the official PEI-US 2016 codebook or questionnaire. In particular, exact definitions of individual CSV columns should be verified against the official dataset documentation. The RAG system should eventually contain the official codebook/methodology as its primary knowledge source.

16. Recommended Knowledge Files

For a stronger RAG system, the documents folder should eventually contain:

- This dataset guide
- Official PEI-US 2016 codebook
- Official questionnaire/survey documentation
- PEI-US 2016 methodology/report
- Any official variable-definition or technical documentation available with the dataset.

17. Source Note

The factual background in this guide is based on information published by the Electoral Integrity Project concerning the PEI-US 2016 study, including its scope, expert survey, 726 responses, 49 core items, and 100-point PEI Index. This project guide intentionally avoids inventing exact definitions for individual CSV columns that have not been verified against the official codebook.